

COMPSCI 589 Final Project - Spring 2026

Due **May 8**, 11:55 pm Eastern Time

Introduction

- Each group member was responsible for evaluating the performance of at least one unique algorithm implemented during the semester on each dataset
- The algorithms tested in this project were chosen from the pool of: k -NN, Decision Trees, multinomial Naive Bayes, Random Forests, and Neural Networks
- All work was done in the shared github repository https://github.com/19851xw/589_final_project

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- **Katherine** implemented TODO
 - **Tiffany** implemented TODO
 - **Xing-Wei** implemented Decision Trees

Experiments and Analyses

1 The Hand-Written Digits Recognition Dataset

1.1 Neural Networks Algorithm

TODO

1.2 Algorithm 2

TODO

1.3 Decision Trees Algorithm

Tuning for dataset with 1797 rows and 64 features.

Testing Depths: [7, 9, 11, 13]

Testing Splits: [17, 53, 89, 179]

Depth	Split	Test Acc	Test F1
7	17	0.8136	0.8124
7	53	0.7561	0.7568
7	89	0.7062	0.7043
7	179	0.6328	0.6165
9	17	0.8319	0.8311
9	53	0.7612	0.7592
9	89	0.7067	0.7013
9	179	0.6505	0.6360
11	17	0.8219	0.8222
11	53	0.7626	0.7615
11	89	0.7204	0.7171
11	179	0.6321	0.6216
13	17	0.8141	0.8119
13	53	0.7568	0.7568
13	89	0.7039	0.7037
13	179	0.6383	0.6217

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BEST CONFIGURATION:

Max Depth: 9

Min Split: 17

F1 Score: 0.8311

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Final Results:

Training Accuracy: 0.8936

Training F1 Score: 0.8935

Testing Accuracy: 0.8344

Testing F1 Score: 0.8327

Analysis about graph

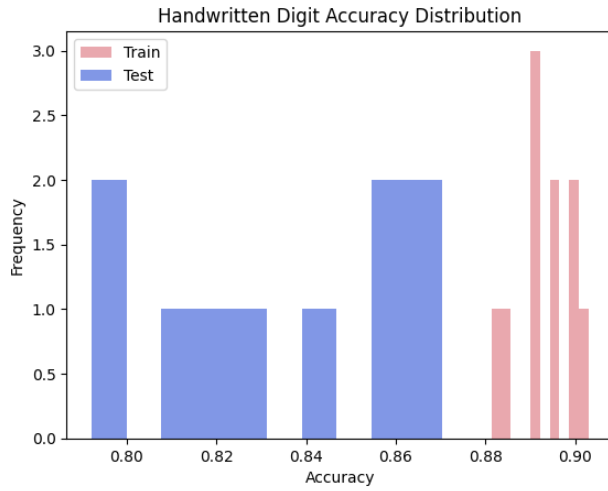


Figure 1: Handwritten Digits Accuracy Distribution

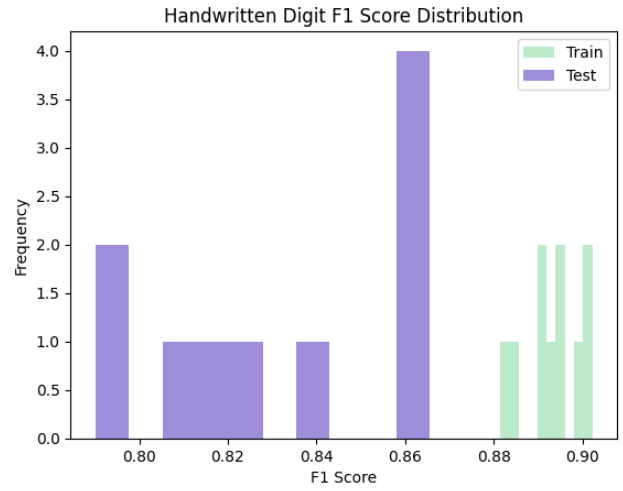


Figure 2: Handwritten Digits F1 Score Distribution

2 The Oxford Parkinson's Disease Detection Dataset

2.1 Neural Networks Algorithm

TODO

2.2 Algorithm 2

TODO

2.3 Decision Trees Algorithm

Tuning for dataset with 195 rows and 22 features.

Testing Depths: [2, 3, 5, 7, 9, 11, 13]

Testing Splits: [9, 19, 29, 39]

Depth	Split	Test Acc	Test F1
2	9	0.8503	0.8065
2	19	0.8117	0.7631
2	29	0.8454	0.7922
2	39	0.8625	0.7836
3	9	0.8681	0.8278
3	19	0.8312	0.7890
3	29	0.8309	0.7553
3	39	0.8620	0.7975
5	9	0.8575	0.8157
5	19	0.8556	0.8171
5	29	0.8362	0.7784
5	39	0.8611	0.7906
7	9	0.8456	0.8050
7	19	0.8211	0.7456
7	29	0.8312	0.7815
7	39	0.8617	0.7912
9	9	0.8323	0.7762

9	19	0.8515	0.8086
9	29	0.8534	0.8038
9	39	0.8512	0.7767
11	9	0.8562	0.7975
11	19	0.8220	0.7627
11	29	0.8273	0.7595
11	39	0.8606	0.7840
13	9	0.8628	0.8177
13	19	0.8361	0.7720
13	29	0.8348	0.7692
13	39	0.8495	0.7635

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BEST CONFIGURATION:
Max Depth: 3
Min Split: 9
F1 Score: 0.8278
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Final Results:
Training Accuracy: 0.9345
Training F1 Score: 0.9165
Testing Accuracy: 0.8508
Testing F1 Score: 0.8101

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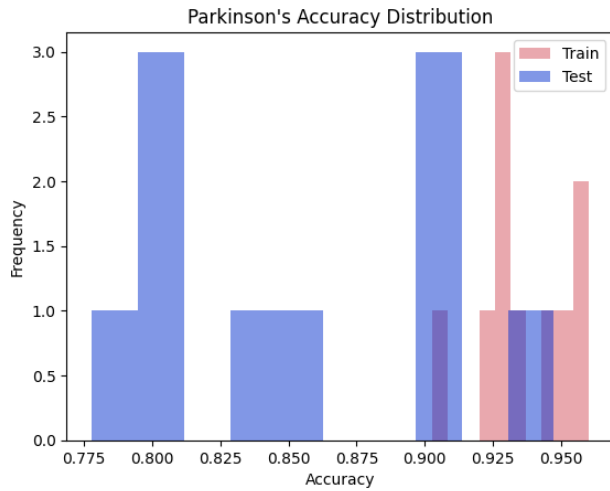


Figure 3: Oxford Parkinson's Disease Detection Accuracy Distribution

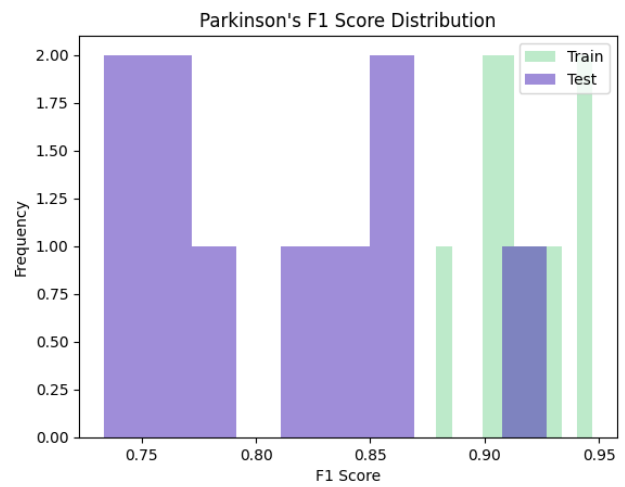


Figure 4: Oxford Parkinson's Disease Detection F1 Score Distribution

3 The Rice Grain Dataset

3.1 Neural Networks Algorithm

TODO

3.2 Algorithm 2

TODO

3.3 Decision Trees Algorithm

We chose to test decision trees on the rice dataset because it has a moderate number of features (7) and a relatively large number of samples (3810), which allows us to explore how different tree depths and minimum split sizes affect performance.

Tuning for dataset with 3810 rows and 7 features.

Testing Depths: [2, 3, 5, 7, 9, 11]

Testing Splits: [190, 381, 571, 762]

Depth	Split	Test Acc	Test F1
2	190	0.9215	0.9192
2	381	0.9228	0.9205
2	571	0.9226	0.9202
2	762	0.9239	0.9215
3	190	0.9234	0.9210
3	381	0.9213	0.9189
3	571	0.9202	0.9179
3	762	0.9239	0.9215
5	190	0.9213	0.9189
5	381	0.9234	0.9210
5	571	0.9226	0.9202
5	762	0.9239	0.9215
7	190	0.9223	0.9199
7	381	0.9226	0.9203
7	571	0.9215	0.9192
7	762	0.9239	0.9215
9	190	0.9234	0.9210
9	381	0.9220	0.9197
9	571	0.9239	0.9215
9	762	0.9239	0.9215
11	190	0.9223	0.9199
11	381	0.9220	0.9197
11	571	0.9218	0.9194
11	762	0.9239	0.9215

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BEST CONFIGURATION:

Max Depth: 11

Min Split: 762

F1 Score: 0.9215

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Final Results:

Training Accuracy: 0.9239

Training F1 Score: 0.9216

Testing Accuracy: 0.9239

Testing F1 Score: 0.9215

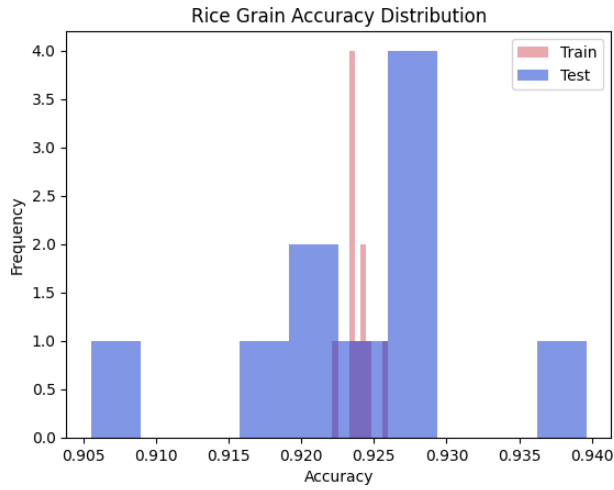


Figure 5: Rice Accuracy Distribution

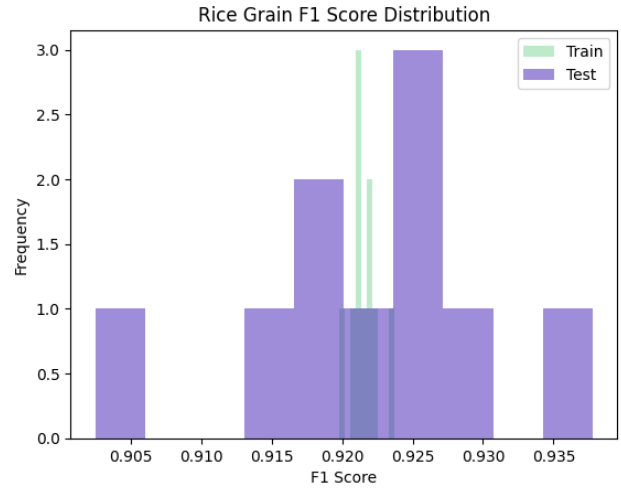


Figure 6: Rice F1 Score Distribution

For both models, the training scores are highly concentrated around 0.92, while the testing scores are more spread out, with a mean around 0.923 for accuracy and 0.922 for F1 score. The training performance suggests that the models have high stability and low bias, however, the testing performance indicates that there may be some sensitivity to the specific training data used in each fold. Since the mean training scores are very close to the mean testing scores, it suggests that the models are robust and are not overfitting. Decision trees can easily split numerical features to find precise boundaries, so they are well-suited for the rice dataset, which consists of numerical attributes.

4 The Credit Approval Dataset

4.1 Neural Networks Algorithm

TODO

4.2 Algorithm 2

TODO

4.3 Decision Trees Algorithm

Tuning for dataset with 653 rows and 15 features.

Testing Depths: [2, 4, 6, 8, 10, 12]

Testing Splits: [32, 65, 97, 130]

Depth	Split	Test Acc	Test F1
2	32	0.8622	0.8621
2	65	0.8639	0.8636
2	97	0.8622	0.8622
2	130	0.8622	0.8620
4	32	0.8378	0.8358
4	65	0.8482	0.8473
4	97	0.8588	0.8580
4	130	0.8577	0.8566
6	32	0.8318	0.8293
6	65	0.8302	0.8288
6	97	0.8501	0.8491
6	130	0.8610	0.8606
8	32	0.8378	0.8356
8	65	0.8576	0.8568
8	97	0.8496	0.8489
8	130	0.8620	0.8612
10	32	0.8377	0.8356
10	65	0.8409	0.8397
10	97	0.8437	0.8423
10	130	0.8504	0.8495
12	32	0.8375	0.8350
12	65	0.8451	0.8439
12	97	0.8668	0.8654
12	130	0.8635	0.8626

=====

BEST CONFIGURATION:

Max Depth: 12

Min Split: 97

F1 Score: 0.8654

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Final Results:

Training Accuracy: 0.8746

Training F1 Score: 0.8741

Testing Accuracy: 0.8606
Testing F1 Score: 0.8601

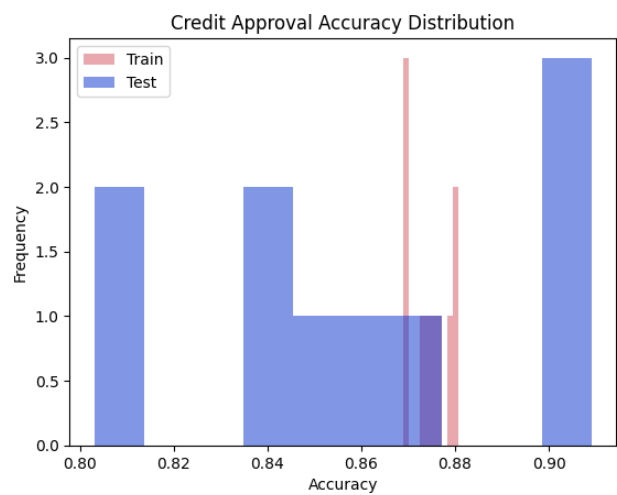


Figure 7: Credit Approval Accuracy Distribution

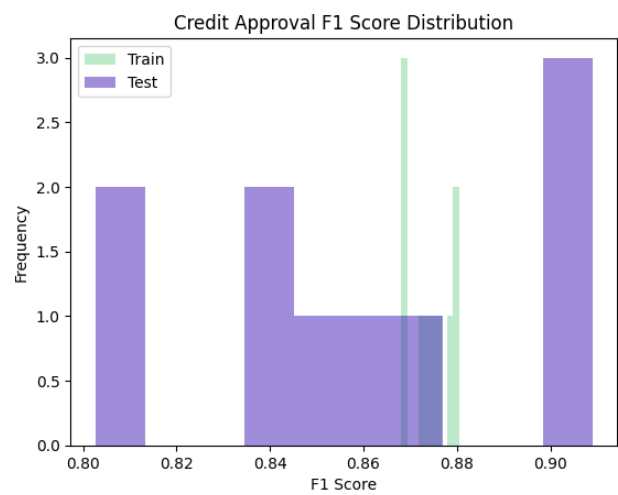


Figure 8: Credit Approval F1 Score Distribution

	Handwritten Digits		Parkinson's		Rice Grains		Credit Approval	
	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score	Accuracy	F1-Score
Neural Networks								
K-Nearest Neighbors								
Decision Trees	0.8344	0.8327	0.8508	0.8101	0.9239	0.9215	0.8606	0.8601

Extra Credit

We did the following extra credit tasks. Describe what you did as part of that task and present the corresponding results. Any source code you develop should be included in your Gradescope submission.

[QE.1] [Extra Credit Question #1] (10 Points)

You may earn up to 10 extra points if you analyze all four datasets using an additional algorithm. In other words, you would be evaluating more than two algorithms on each of the datasets discussed in Section 1.

[QE.2] [Extra Credit Question #2] (10 Points)

You may earn up to 10 extra points if you select a new challenging dataset and evaluate the performance of different algorithms on it. A challenging dataset should necessarily include both categorical and numerical attributes and have more than two classes; or it should be a dataset where you have to classify images.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you combine different algorithms and construct an ensemble. For instance, you could create an ensemble composed of three neural networks (each with a different architecture) and a random forest. The ensemble would be trained as described in class: each algorithm would be trained on its own bootstrap dataset, and so on. The final predictions would then be determined by majority voting. Evaluate your ensemble algorithm on all four datasets discussed in Section 1.

[QE.3] [Extra Credit Question #3] (15 Points)

You may earn up to 15 extra points if you implement a new type of algorithm—a non-trivial variant of one of the algorithms studied in class. If you choose to do so, you should evaluate its performance on all four datasets.